

Krebs Cycle

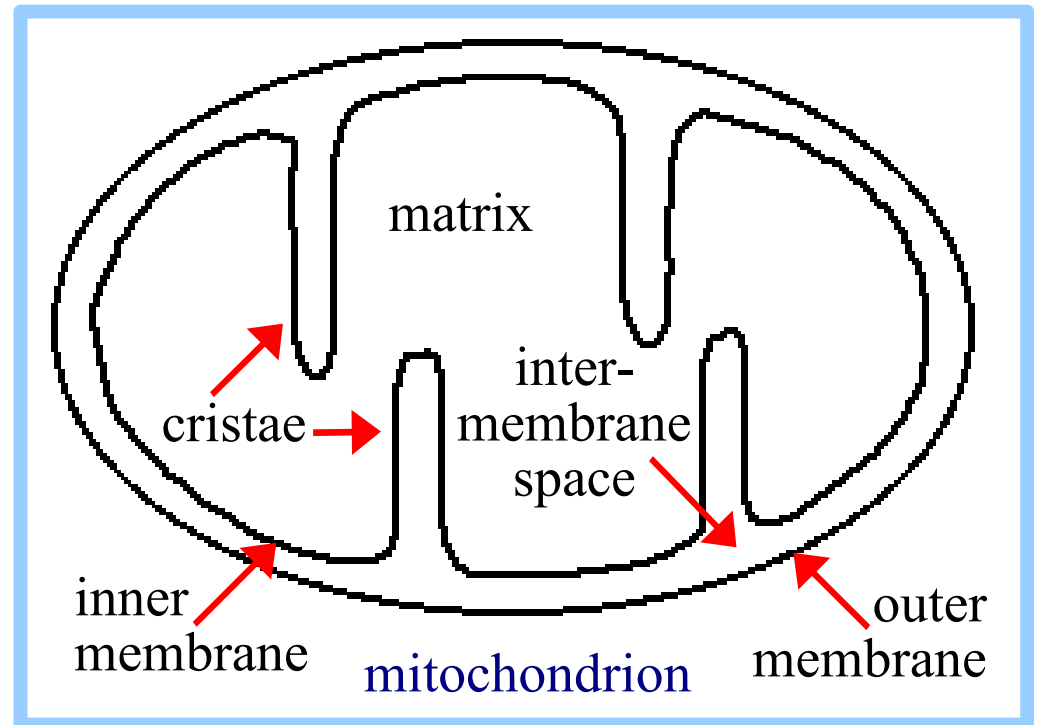
CAC Introduction

I) Localization

Glycolysis happens in the cell's cytosol

Pyruvate enters the mitochondrion to be metabolized further by the CAC

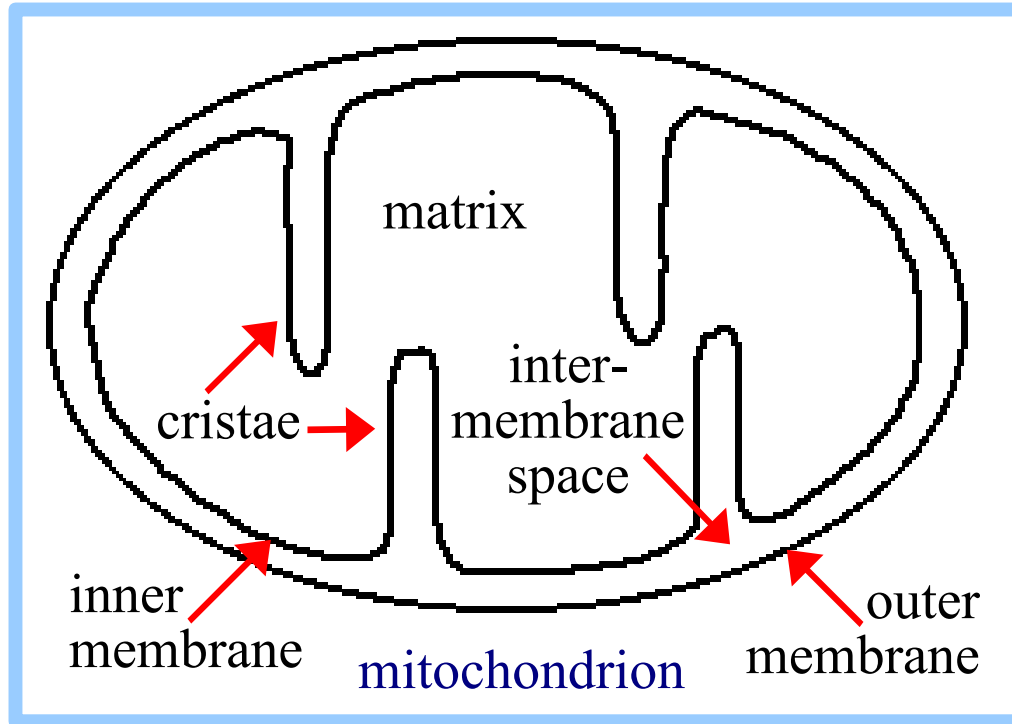
II) Mitochondrial Compartments



■ The matrix contains

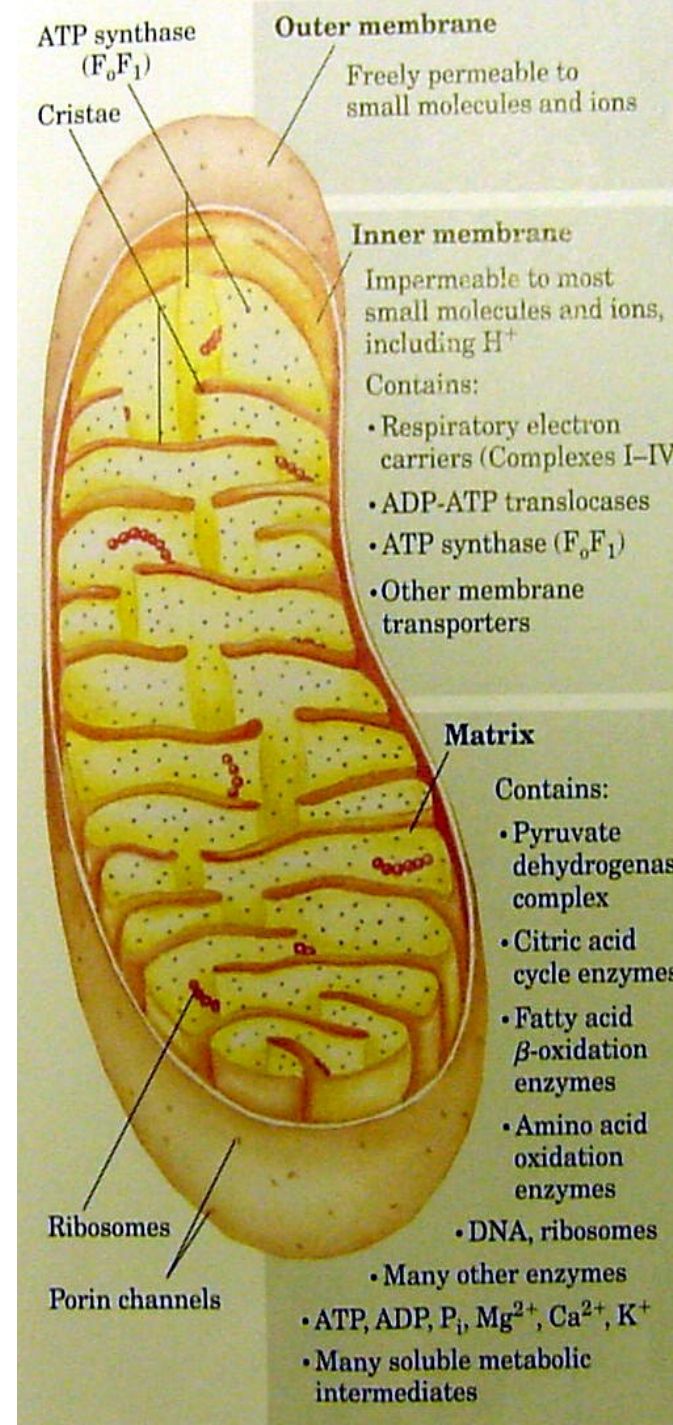
- Pyruvate dehydrogenase complex,
- enzymes of the CAC, and other pathways, i.e. fatty acid oxidation & amino acid metabolism.
- The outer membrane contains large channels and is “leaky” to ions & small molecules

CAC Introduction



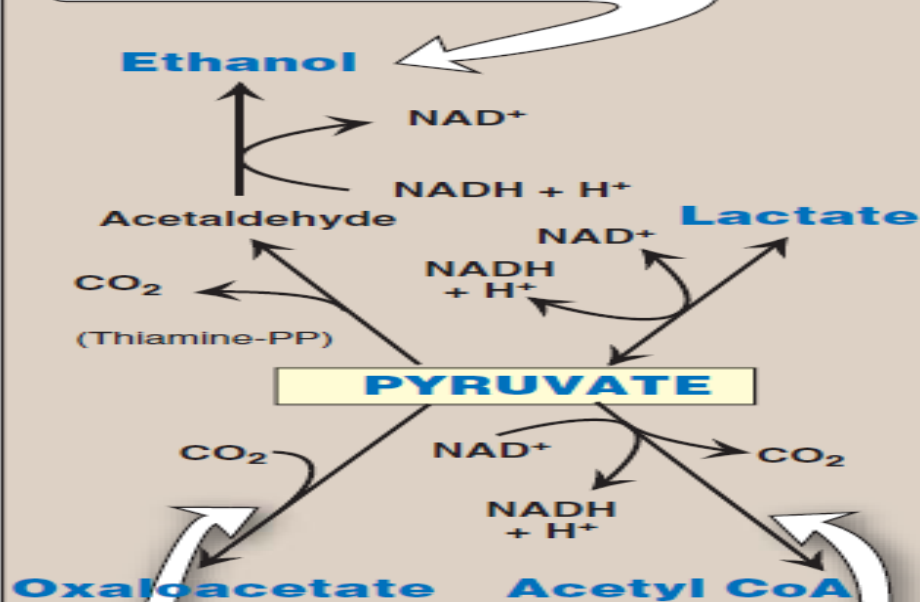
■ Inner membrane infoldings, called cristae,

- contain parts of the respiratory chain & F_1F_0 ATP Synthase and is the major permeability barrier
- It contains various transporters, including a carrier protein that allows pyruvate to enter the matrix



ETHANOL SYNTHESIS

- Occurs in yeast and some bacteria (including intestinal flora).
- Thiamine pyrophosphate-dependent pathway.



PYRUVATE DEHYDROGENASE COMPLEX

- Inhibited by acetyl CoA.
- Source of acetyl CoA for TCA and fatty acid synthesis.
- An irreversible reaction.

PYRUVATE CARBOXYLASE

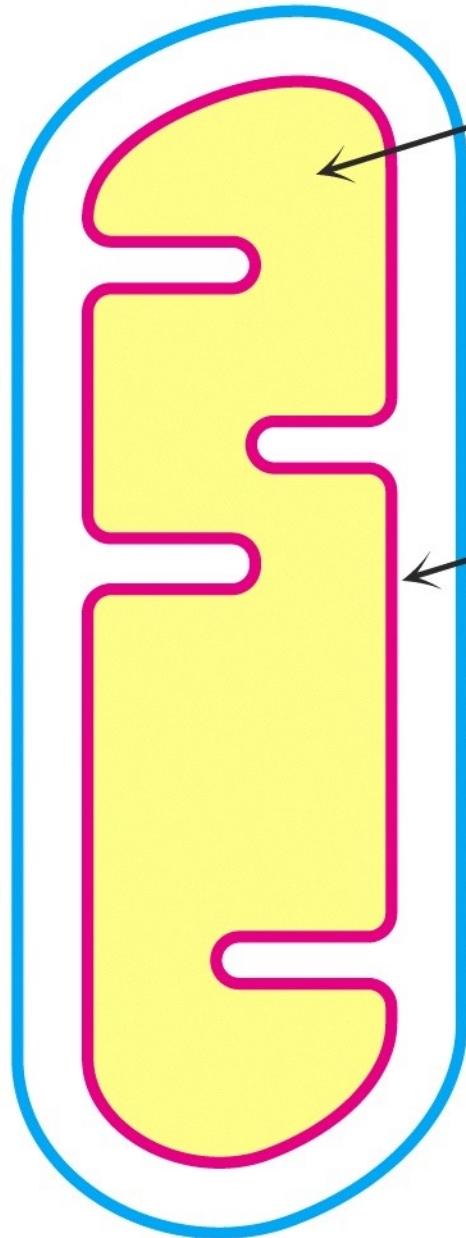
- Activated by acetyl CoA.
- Replenishes intermediates of the TCA cycle.
- Provides substrates for gluconeogenesis.
- An irreversible reaction.

Fig. Summary of the metabolic fates of pyruvate.

Pyruvate transport into mitochondria

- Pyruvate generated in the cytoplasm is transported across the inner mitochondrial membrane via a **pyruvate/H⁺ symport**.
- This transport uses some of the energy stored in the mitochondrial inner membrane electrical potential gradient.

Mitochondrion



Matrix

Oxidative decarboxylation of pyruvate, & citric acid cycle take place in matrix, along with fatty acid oxidation

Inner mitochondrial membrane

Site of oxidative phosphorylation

Outer mitochondrial membrane

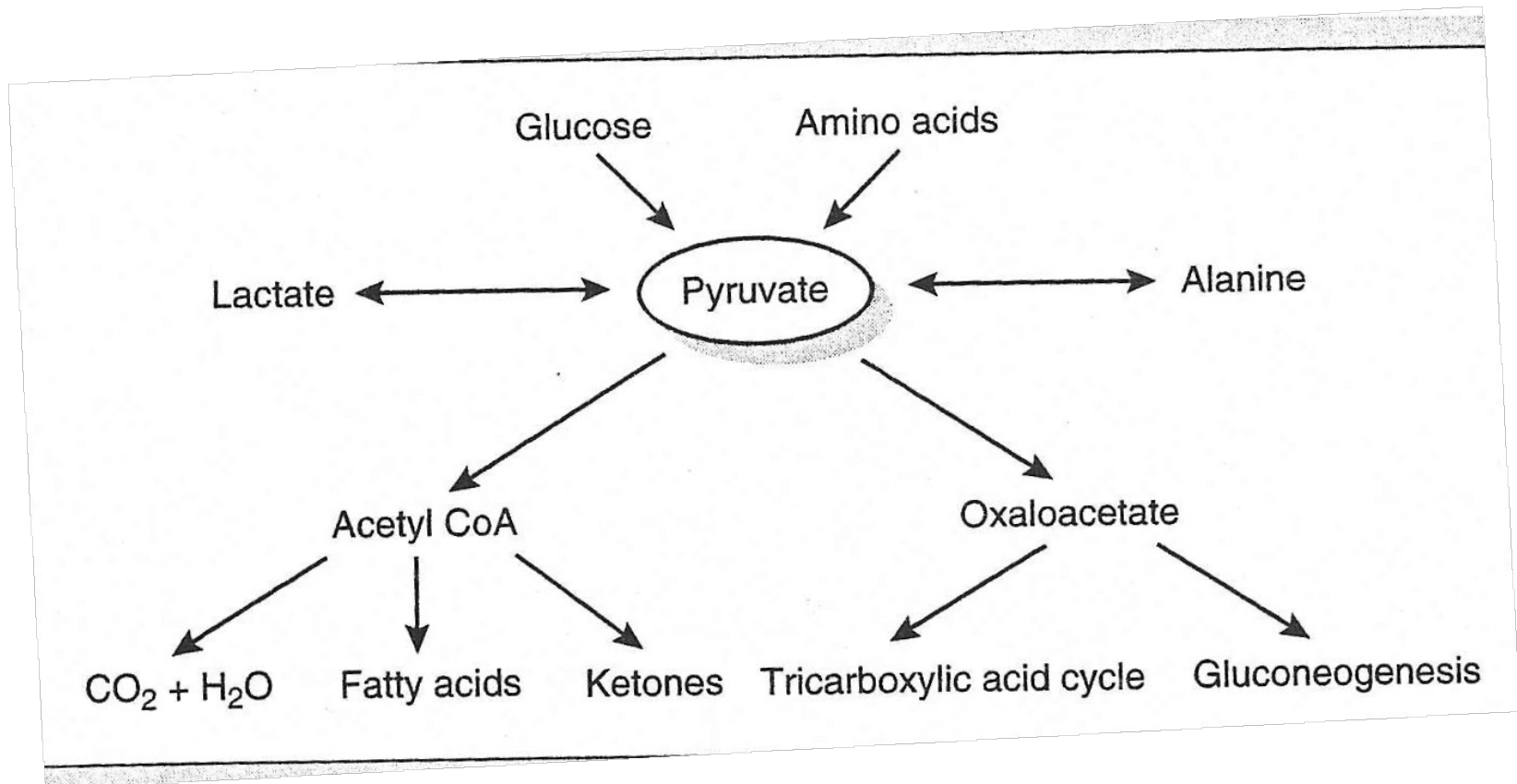
Permeable

Krebs Cycle: Preparatory Step

- Occurs in the **mitochondrial matrix** and is fueled by pyruvic acid and fatty acids
- Pyruvic acid from glycolysis is converted to acetyl coenzyme A (A-CoA) in three main steps:
 - **Decarboxylation**
 - 1 carbon is removed from pyruvic acid; $3C \rightarrow 2C$ molecule
 - The lost carbon forms carbon dioxide; exhaled
 - **Oxidation**
 - 2 Hydrogen atoms are removed from pyruvic acid ('oxidation') and picked up by NAD^+
 - NAD^+ is reduced to $NADH + H^+$.
 - **Formation of acetyl CoA**
 - the resulting acetic acid is combined with coenzyme A, a sulfur-containing coenzyme, to form acetyl CoA (ACoA).
- The pyruvate dehydrogenase complex ***is not part of the TCA cycle proper***, but is a major source of acetyl CoA—the two-carbon substrate for the cycle.

Pyruvate Dehydrogenase:

- The enzyme that links glycolysis with other pathways



The PDH Complex

- Multi-enzyme complex
 - Three enzymes
 - 5 co-enzymes
 - Allows for efficient direct transfer of product from one enzyme to the next

<i>Enzymes</i>	<i>Cofactors</i>	<i>Role in Overall Reaction of PDH Complex</i>
E ₁ (pyruvate dehydrogenase)	Thiamine pyrophosphate	Decarboxylation
E ₂ (dihydrolipoyl transacetylase)	Lipoic acid CoA-SH	Oxidation Acyl transfer
E ₃ (dihydrolipoyl dehydrogenase)	FAD NAD ⁺	Regeneration of lipoic acid
PDH kinase		Phosphorylation and inactivation of E ₁
PDH phosphatase		Dephosphorylation and activation of E ₁

Irreversible Oxidative decarboxylation

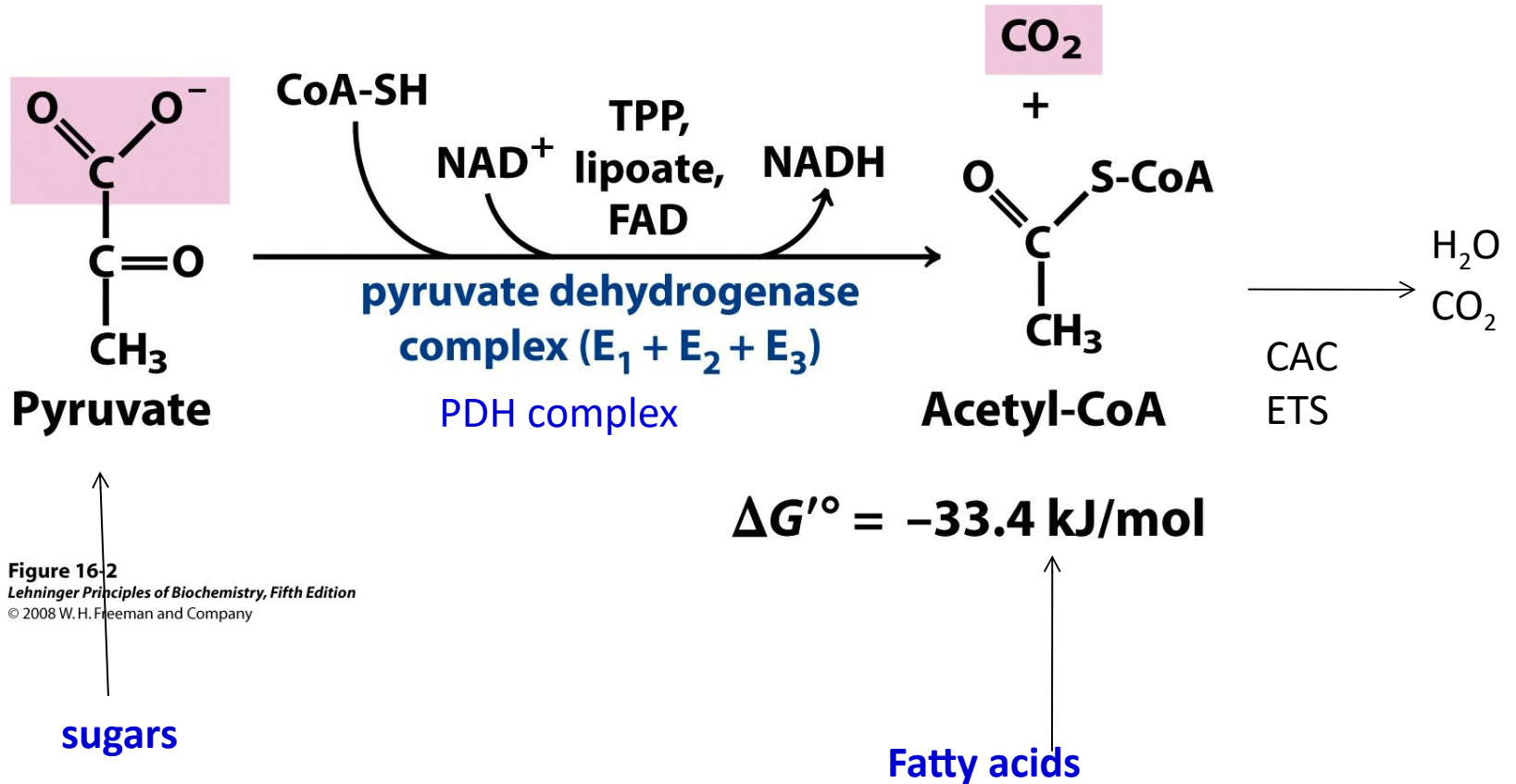


Figure 16-2
Lehninger Principles of Biochemistry, Fifth Edition
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What drives this process?

PDH Complex

Overall

- Acetyl group generated from pyruvate is transferred to coenzyme A
- The final electron acceptor is $\text{NAD}^+ \rightarrow \text{NADH}$
- Acetyl CoA is central in metabolism – it can easily donate acetate based on its “high energy” **thioester linkage**
 - input to CAC (acetate $\rightarrow \text{CO}_2$)
 - donates acetate to fatty acid, ketone body, & cholesterol synthesis

Biomedical Implications

- **Mutations** in the genes for the subunits of the **pyruvate dehydrogenase complex**, or a **dietary thiamine deficiency**, can have severe consequences.
- Thiamine-**deficient** animals are **unable to oxidize pyruvate** normally. Particularly important in the **brain**, which usually obtains energy from the aerobic oxidation of glucose. e.g. **Beriberi** is a disease that results from thiamine deficiency and is characterized by loss of neural function.
- **Alcoholics** can also develop thiamine deficiency because much of their dietary intake of calories is **vitamin-free**.
- An **elevated level of blood pyruvate** is often an indicator of defects in pyruvate oxidation.

Key Points of TCA

- The TCA cycle accounts for over **two-thirds** of the ATP generated from fuel oxidation.
- **Acetyl CoA**, generated from fuel oxidation, is the **substrate** for the TCA cycle.
- Acetyl CoA, when oxidized via the cycle, generates **CO₂, reduced electron carriers, and guanosine triphosphate**.
- The reduced electron carriers (**NADH, FAD[2H]**) donate electrons to O₂ via the **ETC**, which leads to ATP generation from oxidative phosphorylation.

...cont'd

- The citric acid cycle acts as **a final common pathway** for the oxidation of carbohydrate, lipids, and protein.
- The citric acid cycle also plays a major role in **gluconeogenesis, transamination, deamination, and lipogenesis**. **The liver** is the only tissue where all of these occur to a significant extent.
- When large numbers of liver cells are damaged or destroyed, in acute hepatitis or cirrhosis, this can have major repercussions on metabolism.

TCA /Citric Acid/KREBS Cycle

The cycle is located in the mitochondria

All cells have a mitochondria except RBCs

This is the Final common pathway of oxidative metabolism

Acetyl coenzyme A condenses with OAA to begin the cycle. Catabolism of CHO, Fats and Proteins provide the acetyl CoA

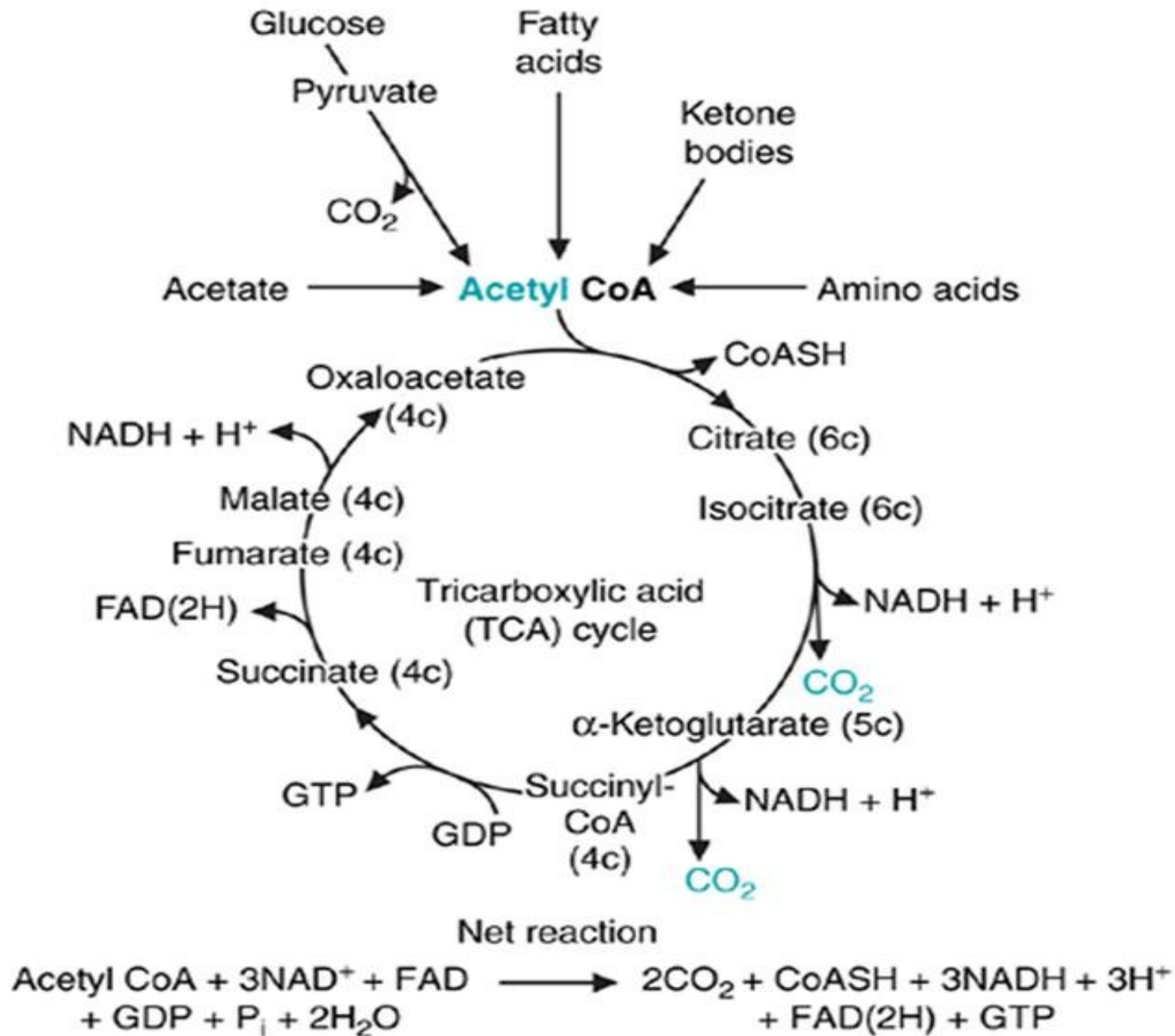
The bulk of ATP used by many cells to maintain homeostasis is produced by the oxidation of pyruvate in the TCA cycle

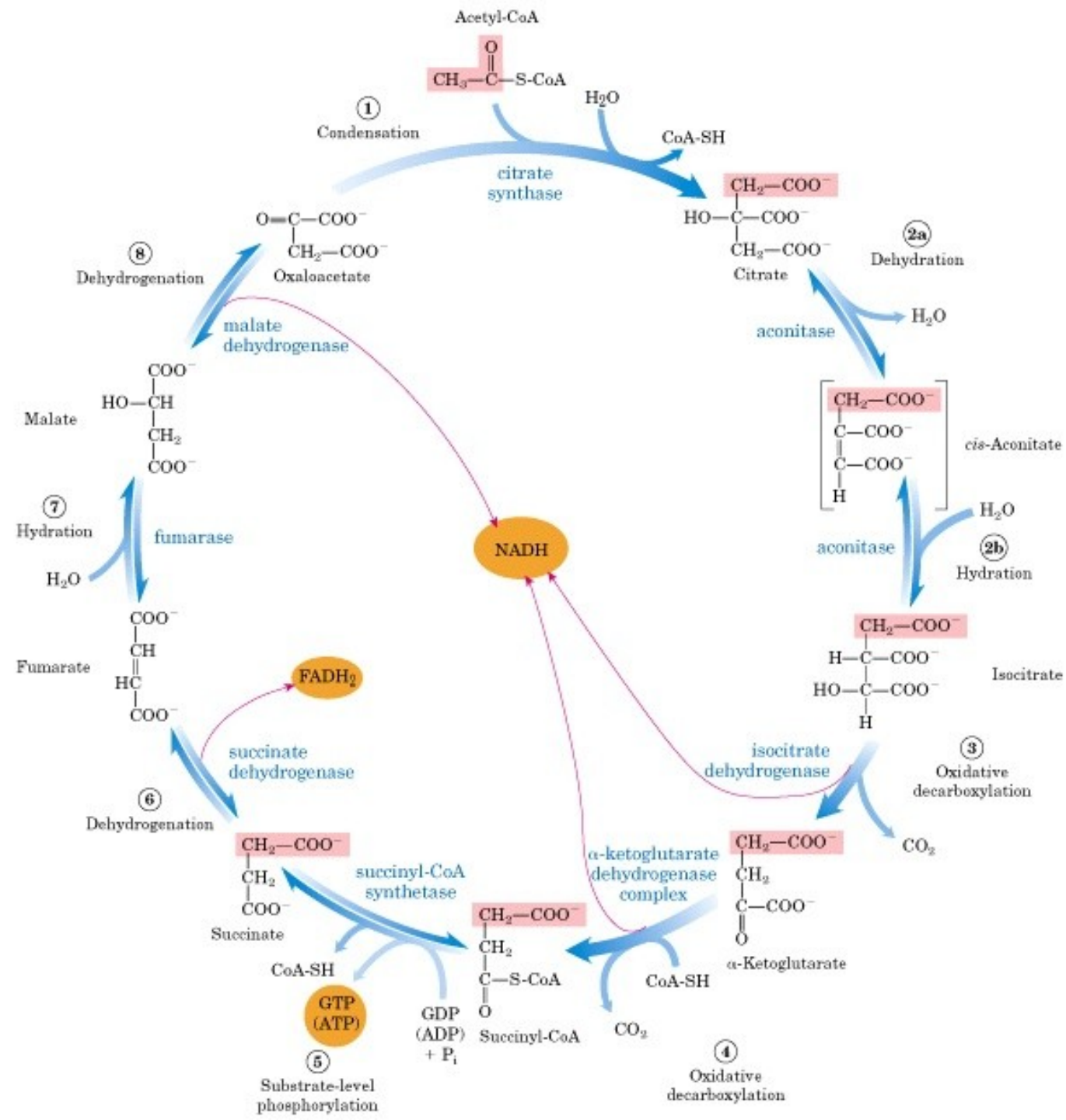
During this oxidation process, reduced **NADH** and reduced **FADH₂** are generated.

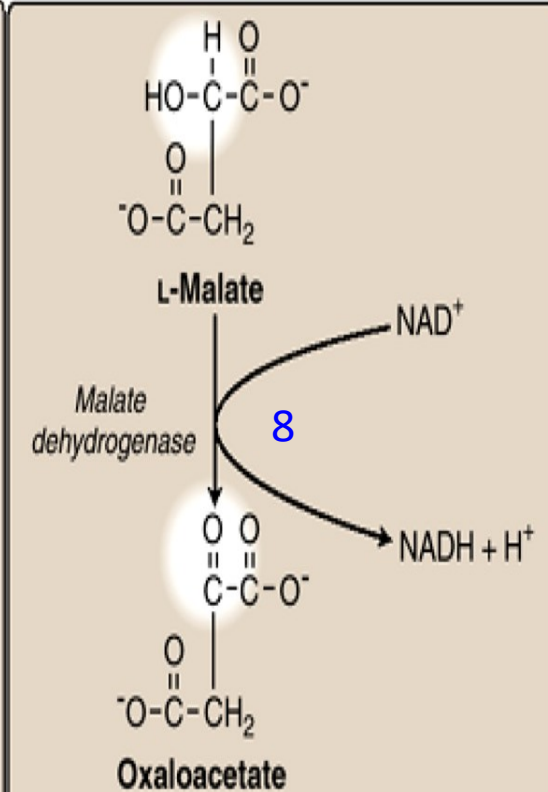
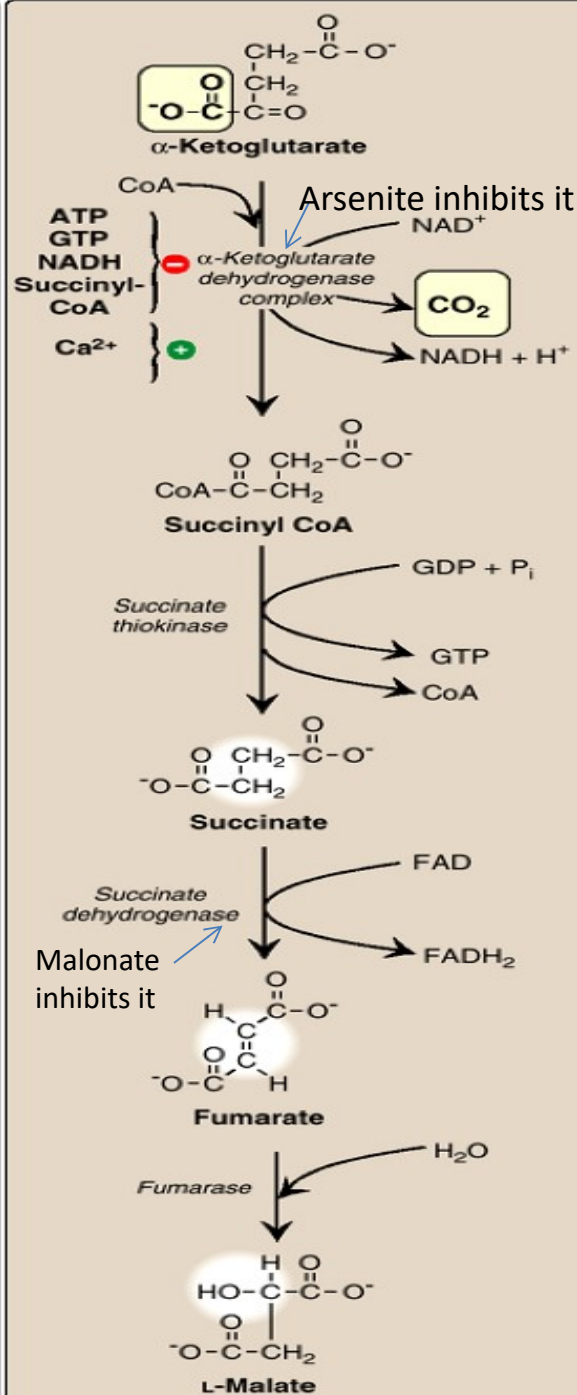
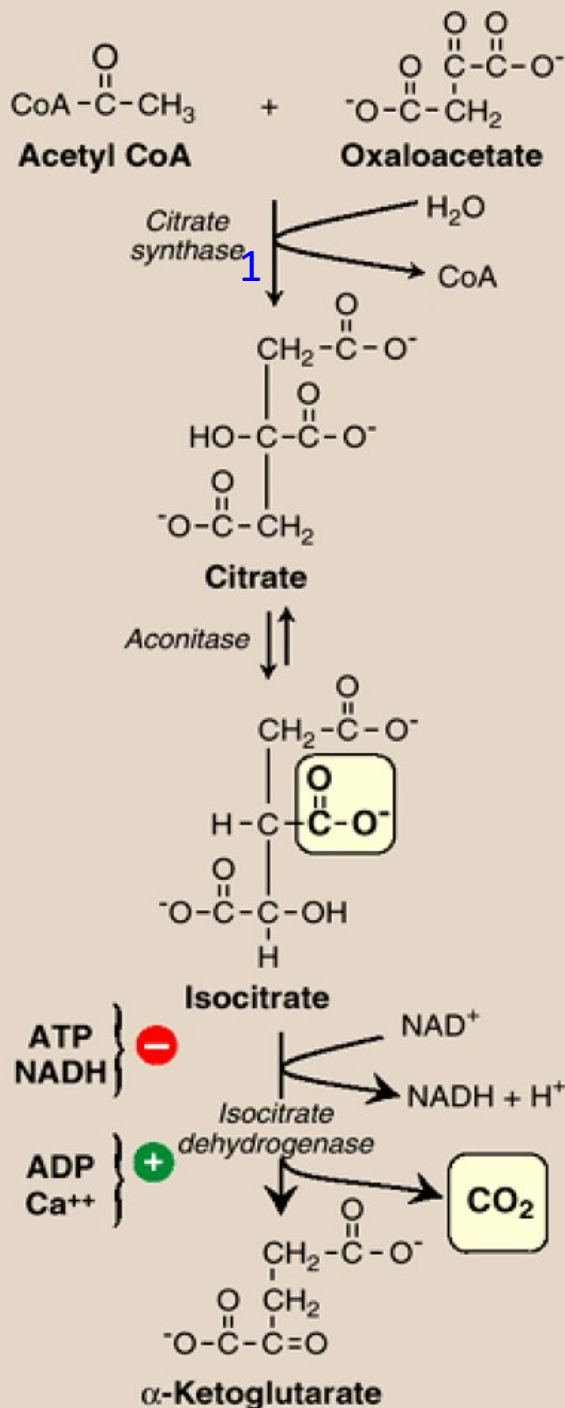
Vitamins play key roles in the citric acid cycle

- Four of the soluble vitamins of the B complex have precise roles in the functioning of the citric acid cycle:
 1. Riboflavin - in the form of flavin adenine dinucleotide (FAD), a cofactor in the dehydrogenase complexes.
 2. Niacin - in the form of nicotinamide adenine dinucleotide (NAD), a cofactor for three dehydrogenases.
 3. Thiamine/vitamin B - as thiamine diphosphate, the coenzyme for the α -ketoglutarate dehydrogenase reaction.
 4. Pantothenic acid - a part of coenzyme A, the cofactor attached in acetyl-CoA and succinyl-CoA

Sources of acetyl-CoA in mitochondria





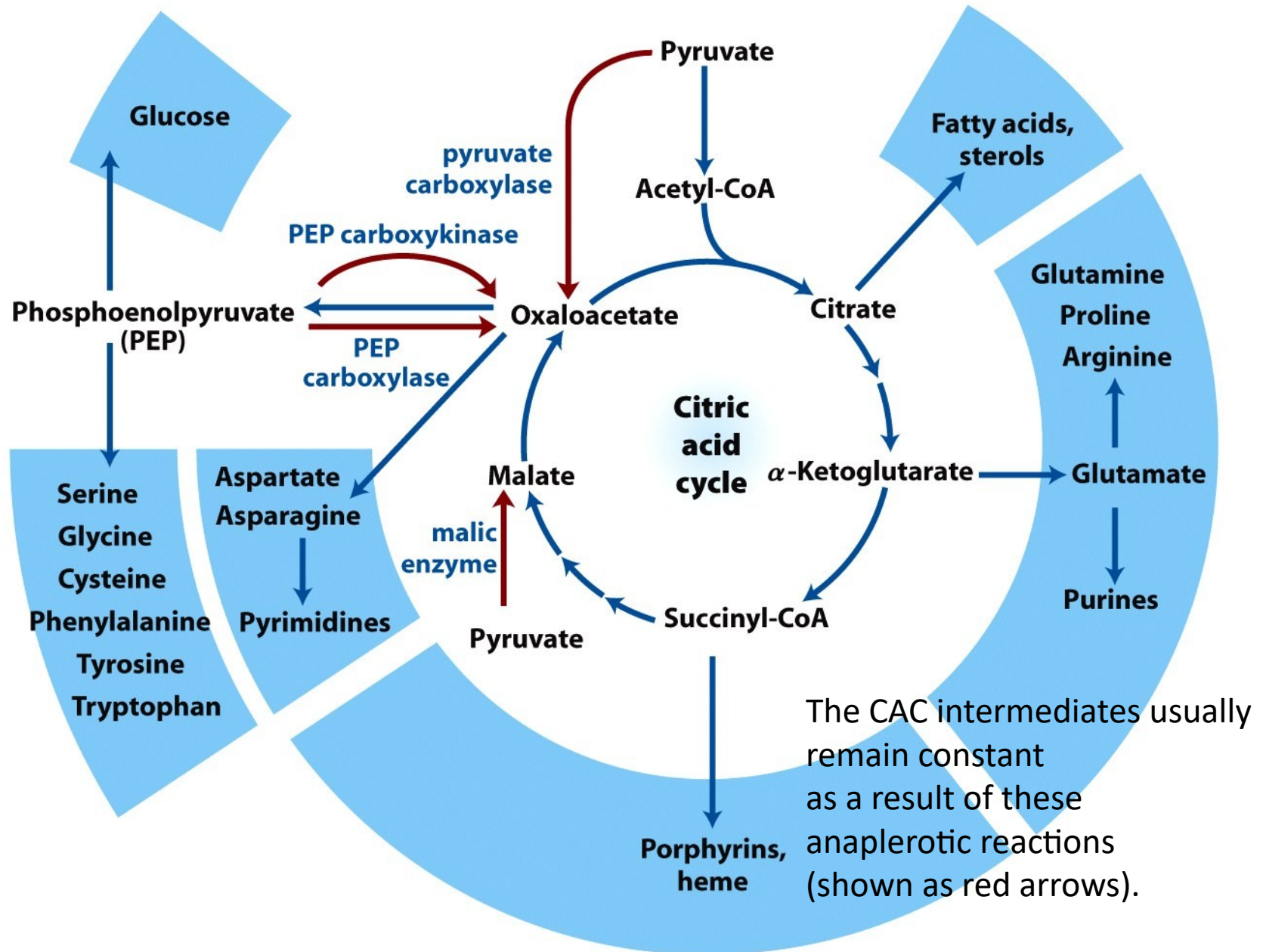


Energy producing reaction	Number of ATP produced
$3 \text{ NADH} \rightarrow 3 \text{ NAD}^+$	9
$\text{FADH}_2 \rightarrow \text{FAD}$	2
$\text{GDP} + \text{P}_i \rightarrow \text{GTP}$	1
12 ATP/acetyl CoA oxidized	

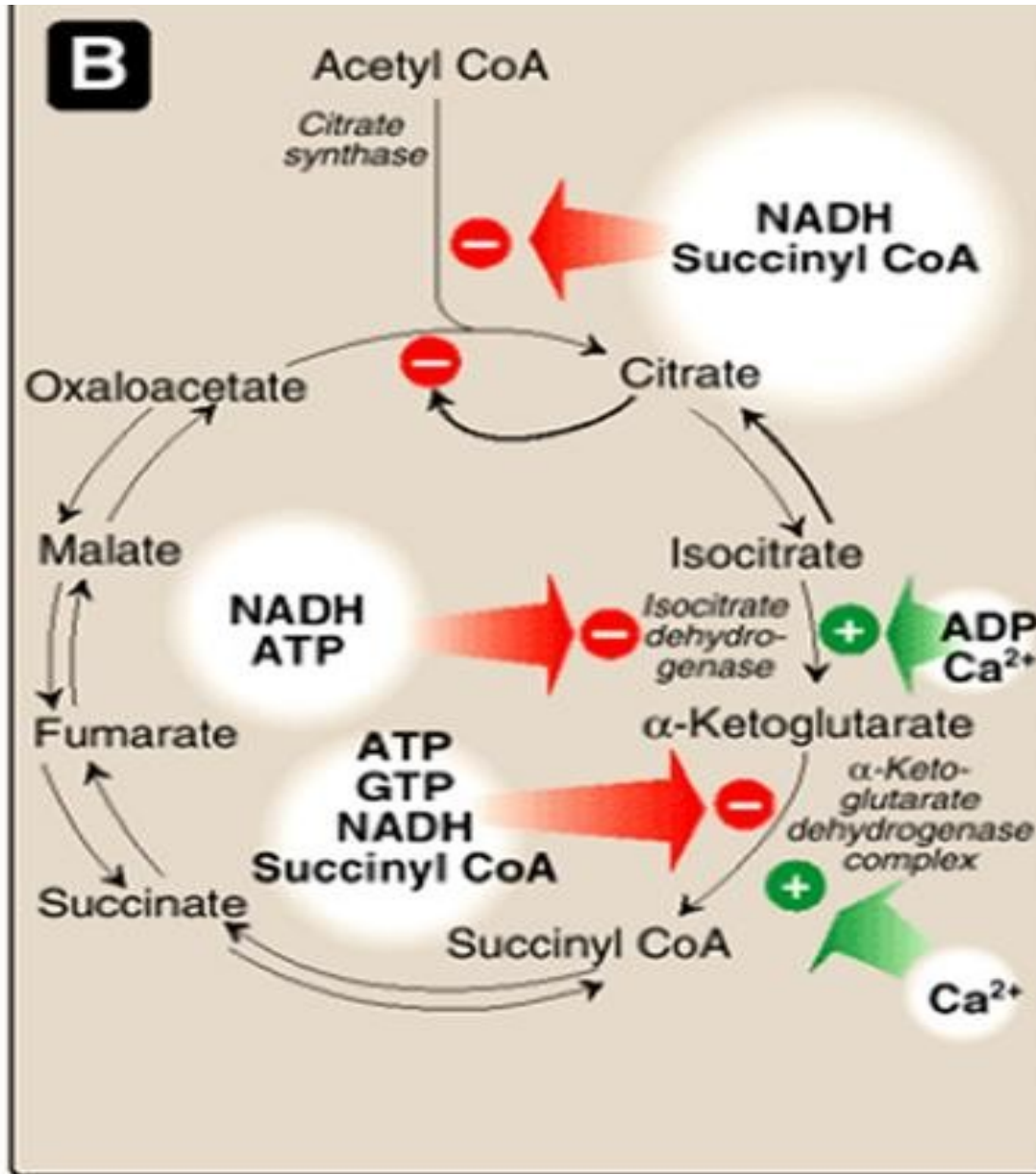
Krebs Cycle

- An eight-step cycle in which each acetic acid is decarboxylated and oxidized, generating:
 - Three molecules of $\text{NADH} + \text{H}^+$ (ox/red)
 - One molecule of FADH_2 (ox/red)
 - Two molecules of CO_2 (decarboxylation)
 - One molecule of ATP (substrate level phosphorylation)
 - For each molecule of glucose entering glycolysis, two molecules of acetyl CoA enter the Krebs cycle
 - Remember, one-6 C Glucose $\rightarrow$ two-2 carbon acetyl coenzyme A (A-CoA)
- 

The Amphibolic Nature of CAC (CAC as “HUB” of the cell)



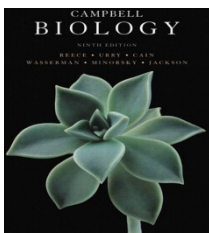
Regulation of the TCA Cycle



The respiratory chain and Oxidative phosphorylation

By the end of this lecture, the student should be able to:

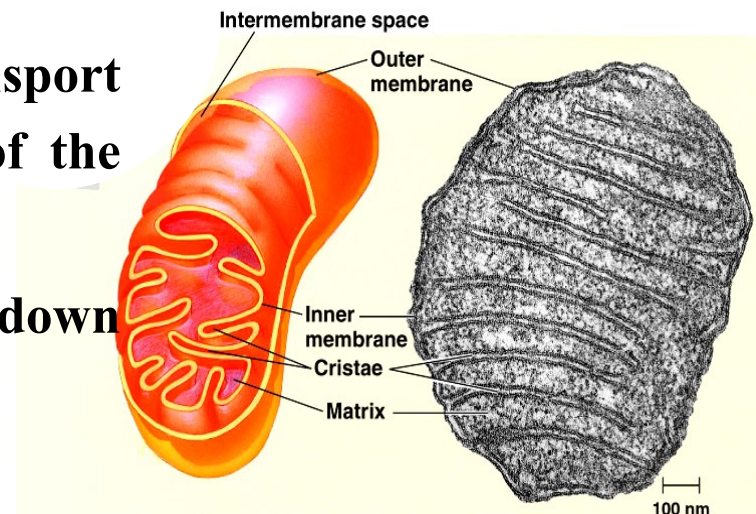
- 1.** The events of the electron transport chain and chemiosmosis.
- 2. Determine the inhibitors of the resp. chain.**
- 3. Understand the concept of shuttles across the mitochondrial membrane.**



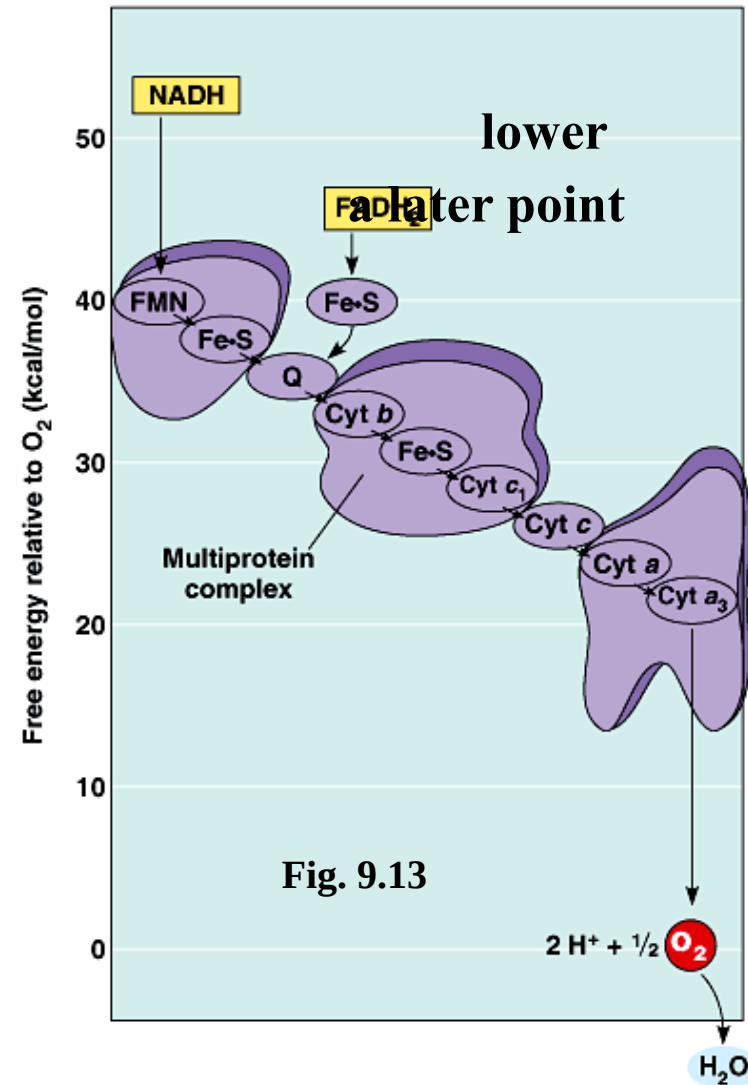
3- Electron transport chain: oxidative phosphorelation

The inner mitochondrial membrane couples electron transport to ATP synthesis (90% of ATP)

- Only 4 of 38 ATP ultimately produced by substrate-level phosphorylation (2 from glycolysis and 2 from Krebs Cycle).
- The vast majority of the ATP (**90%**) comes from the energy in the electrons carried by NADH and FADH₂.
- The energy in these electrons is used in the electron transport chain to power ATP synthesis.
- Thousands of copies of the electron transport chain are found in the extensive surface of the **cristae** (the inner membrane of the mitochondrion).
- Electrons drop in free energy as they pass down the electron transport chain.



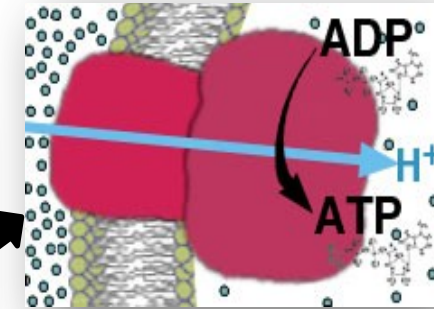
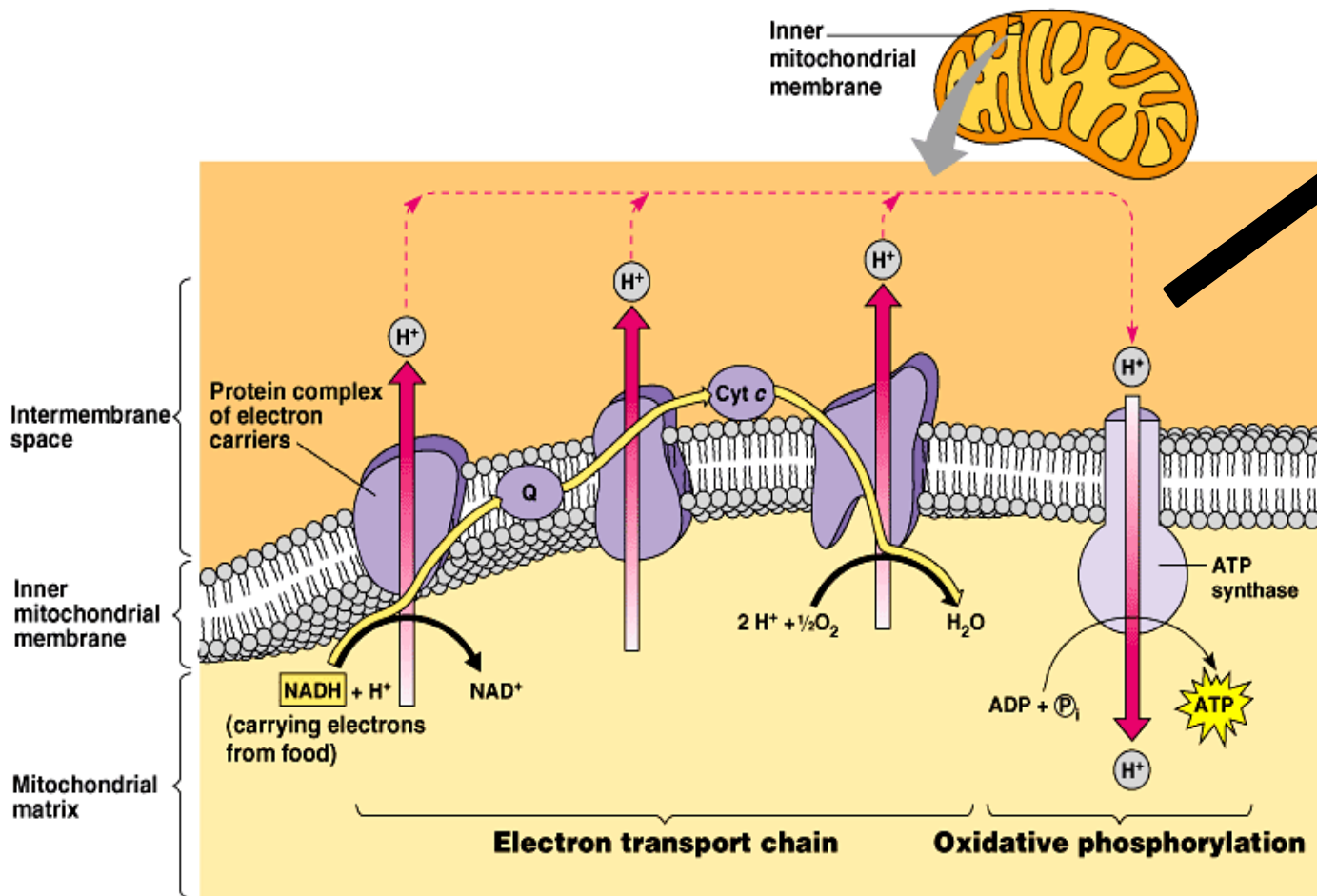
- Electrons carried by **NADH** are transferred to the first molecule in the electron transport chain (the flavoprotein; FMN).
- The electrons continue along the chain which includes several **Cytochrome** proteins and one lipid carrier.
- The electrons carried by **FADH₂** have free energy and are added to in the chain.
- Electrons from **NADH** or **FADH₂** ultimately pass to oxygen.
- The electron transport chain generates no ATP directly. Rather, its function is to break the large free energy drop from food to oxygen into a series of smaller steps that release energy in manageable amounts.



Chemiosmotic Model

How is oxidation coupled to production of ATP?

Peter Mitchell Proposed chemiosmotic hypothesis



1920-1992

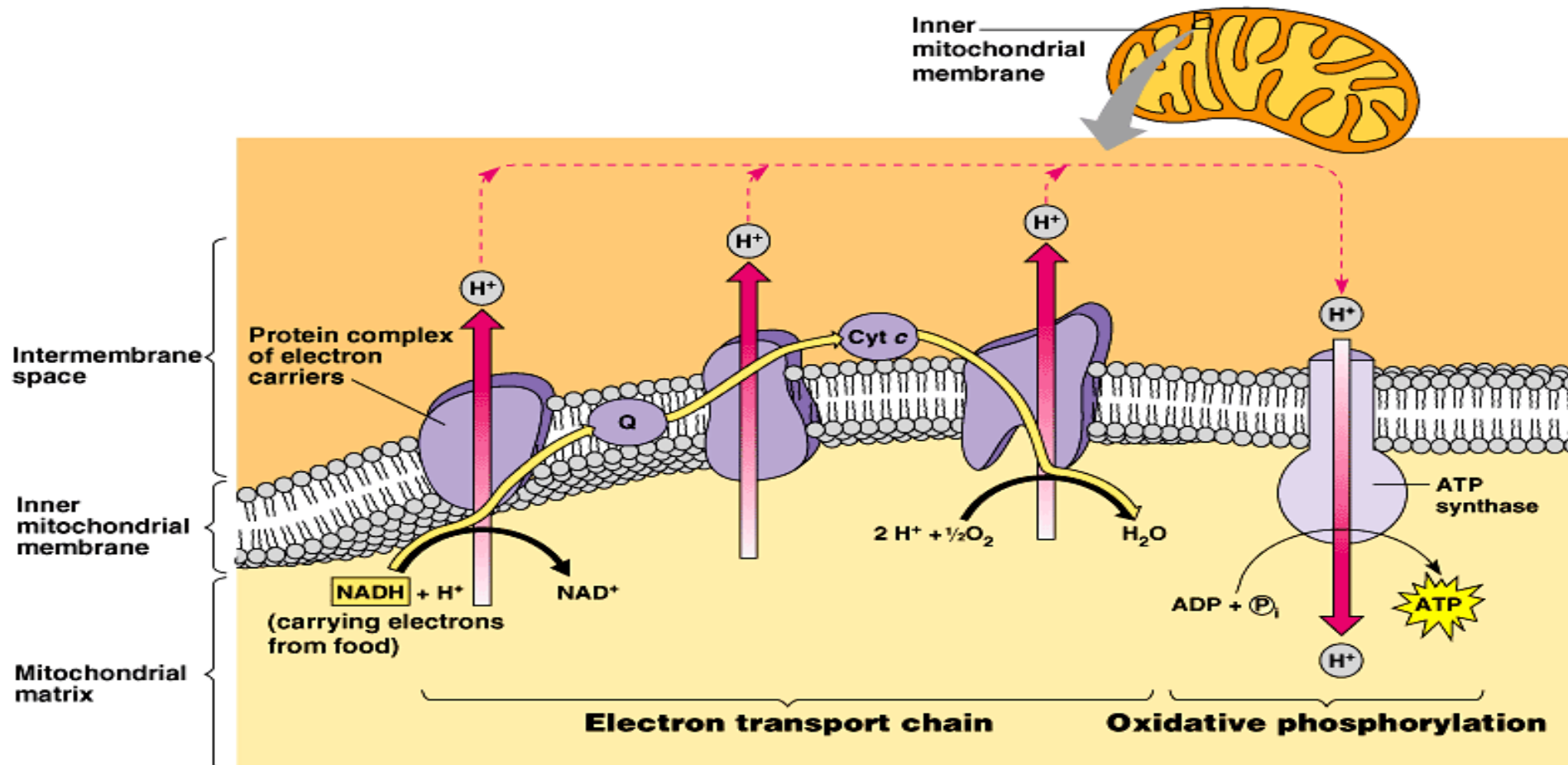


Chemiosmotic Model

The theory postulates that the energy obtained from oxidation is coupled to the translocation of protons H^+ from the mitochondrial matrix to the inter-mitochondrial space.

- Complexes I,III,IV act as proton pumps.**

an **electrochemical gradient** is created across the inner mitochondrial membrane.
(**outside is +ve, inside is negative**)

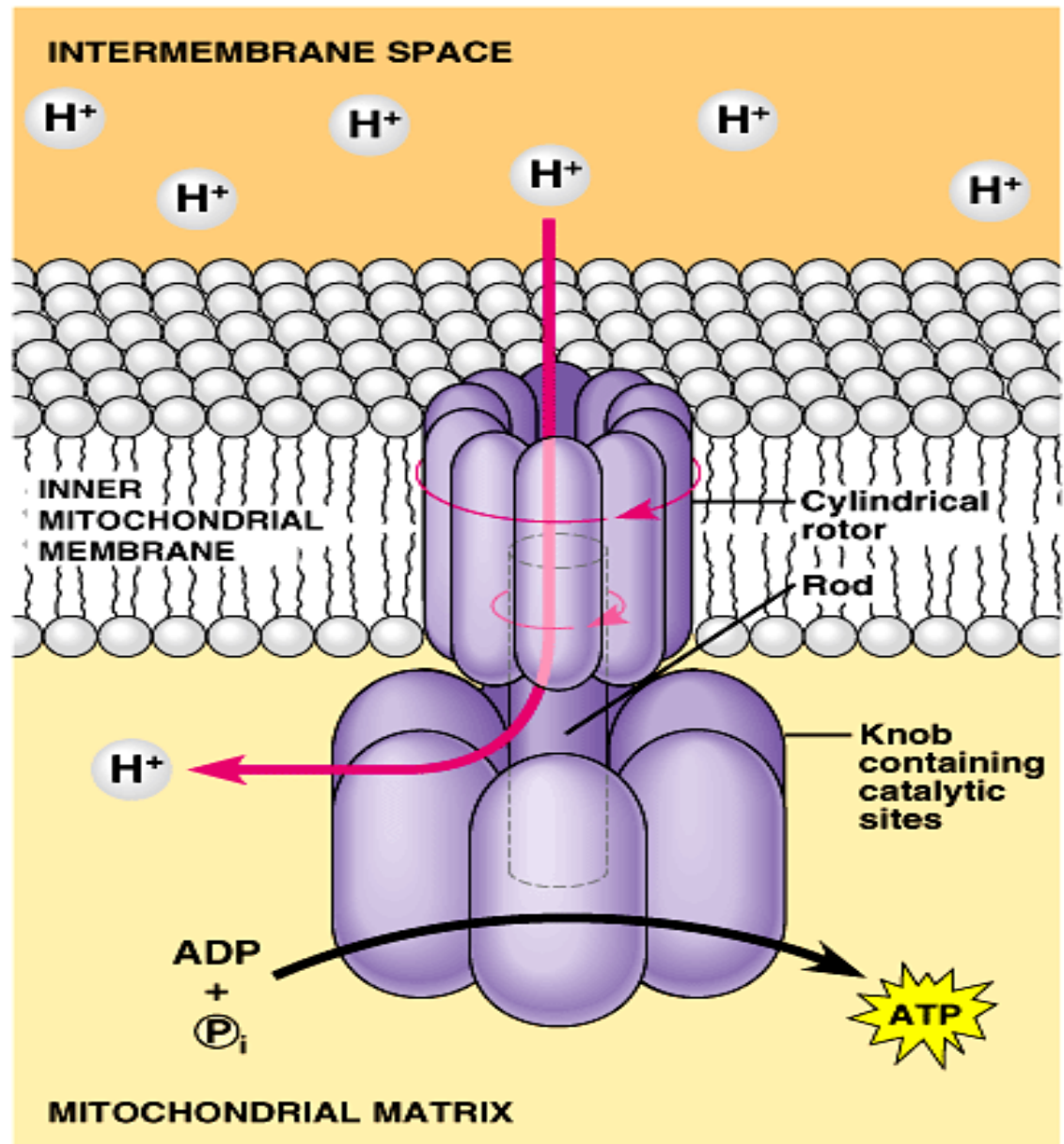


Scattered on the surface of the inner mitochondrial membrane are units of the ATP-synthase.

ATP synthase

F₀

F₁



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Each 2 H⁺ passing through F₀ F₁ gates synthesizes 1 ATP

Chemiosmotic Model

- **Several findings support the chemiosmotic theory:**
 - 1. Addition of protons (acid) to the external medium of mitochondria stimulates ATP production.**
 - 2. Oxidative Phosphorylation does not occur in case of solubilizing mitochondrial membranes.**
 - 3. Uncouplers.**

No. of ATP molecules produced

No. of Oxygen atoms consumed

If we start from NADH = 3:1

If we start from FADH₂ = 2:1

Uncoupler = zero

Respiratory Control

- ⊕ There is no mechanism for storage of ATP.
- ⊕ All ATP present at any moment is only enough to meet the needs of the body for few seconds.
- ⊕ For this reason, there must be an **efficient and controlled way for the production of ATP.**

Respiratory control

1. Mainly by availability of ADP.

(ATP/ADP transporters may be rate limiting at certain times).

2. Availability of electrons:



NADH/NAD ratio or



FADH₂/FAD.

3. Availability of O₂.

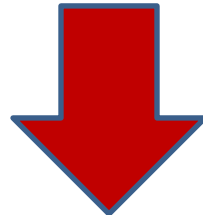
Inhibitors of the respiratory chain

A- Uncouplers

These compounds abolished the coupling between oxidation and phosphorylation through increasing the permeability of the IMM

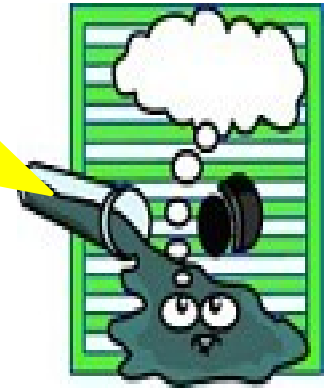


Failure of formation of the electrochemical gradient

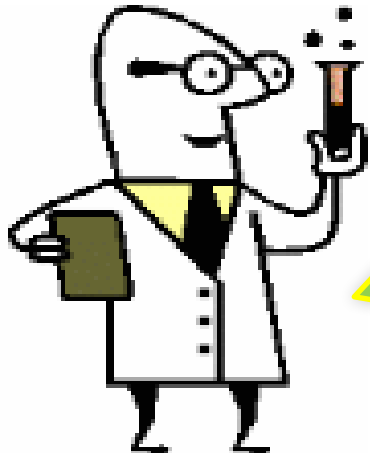


ATP formation stops while oxidation proceeds.

They act by transporting H^+ to inside the mitochondria without passing through F_0F_1
And energy is released as heat.



What is the mechanism of uncouplers?

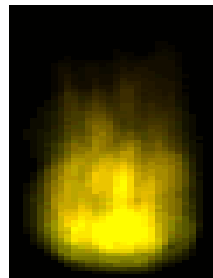


A- Uncouplers

- 1. 2,4 dinitrophenol**
- 2. Dinitrocresol**
- 3. Snake venoms phospholipases**
- 4. Ionophores:** The antibiotic **valinomycin** facilitates the entry of H^+ through the inner mitochondrial membrane.
- 5. Some endogenous compounds when increased:** **Bilirubin and Thyroxine.**

6-Thermogenine (UCP1):

- It is considered as a **physiological uncoupling protein**.
- It is present in the **brown adipose tissue** of newly born, some people and hibernating animals.
- It allows **protons to** pass the mitochondrial matrix without passing F0-F1 complex.
- **No ATP** is formed and energy is released in the form of heat.



Inhibitors of the respiratory chain

B- inhibitors of ETC proper:

- These inhibitors act on the three sites that produce the electrochemical gradient:
 1. **Complex I** is inhibited by **barbiturates** and the insecticide **rotenone**.
 2. **Complex III** is inhibited by the antibiotic, **antimycin A**.
 3. **Complex IV** is inhibited by H_2S , CO and cyanide ( ATP synthesis especially in the CNS causing death).

Inhibitors of the respiratory chain

C-Inhibitors of phosphorylation:

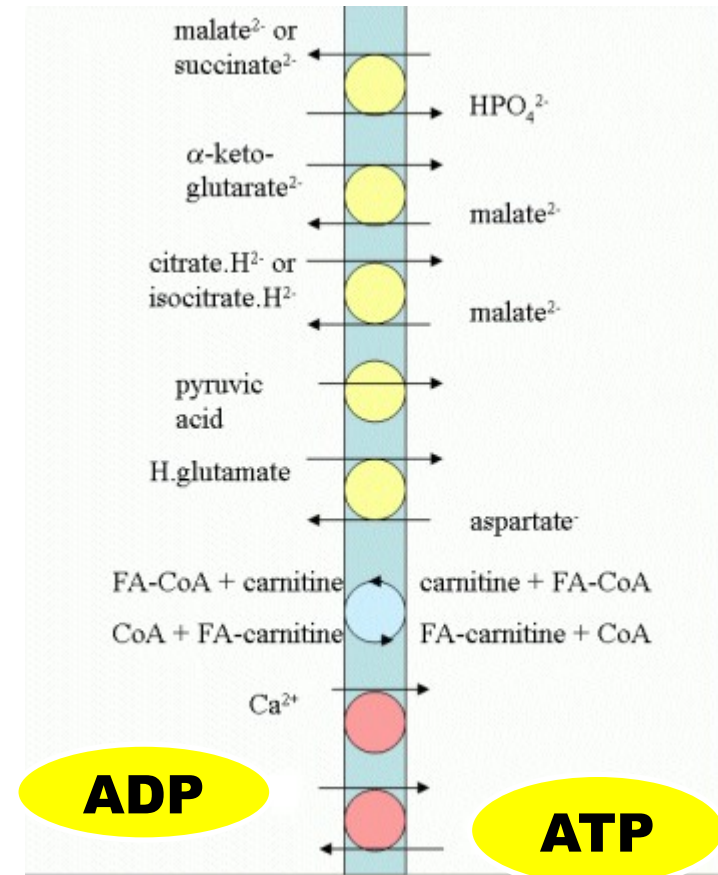
- Completely blocks F_0 so it inhibits ATP synthesis
- Example: **oligomycin (antibiotic)**

D- ATP/ADP transporters inhibitors

- This is achieved by the compound
atractyloside

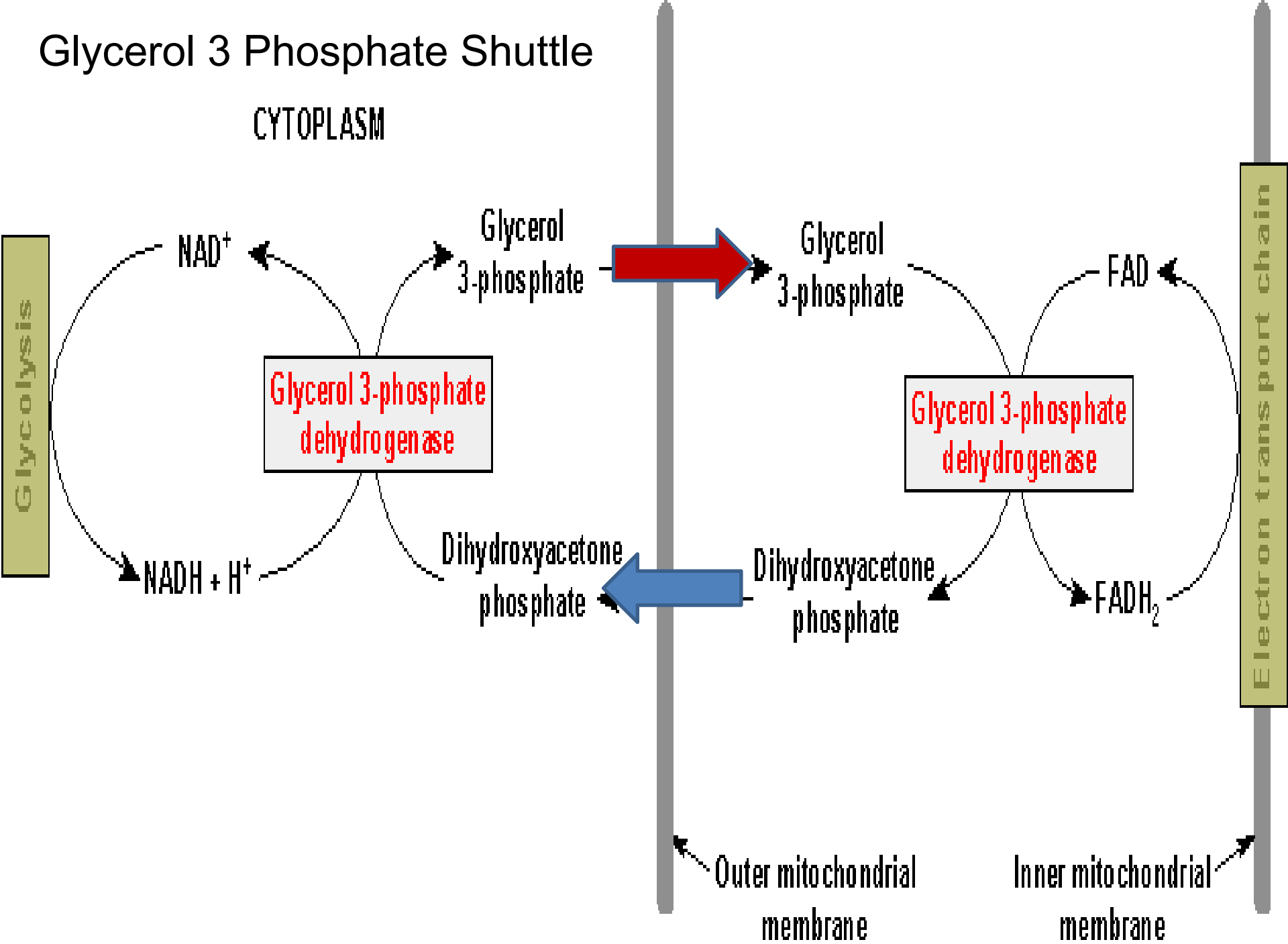
Name	Function	Site of Action
Rotenone	e ⁻ transport inhibitor	Complex I
Amytal	e ⁻ transport inhibitor	Complex I
Antimycin A	e ⁻ transport inhibitor	Complex III
Cyanide	e ⁻ transport inhibitor	Complex IV
Carbon Monoxide	e ⁻ transport inhibitor	Complex IV
Azide	e ⁻ transport inhibitor	Complex IV
2,4,-dinitrophenol	Uncoupling agent	transmembrane H ⁺ carrier
Pentachlorophenol	Uncoupling agent	transmembrane H ⁺ carrier
Oligomycin	Inhibits ATP synthase	ATP synthase

1. **Small molecules** as O_2 , CO_2 , NH_3 and monocarboxylic acids are freely permeable across the membrane.
2. **Di- and tricarboxylic acids** need special transporters.
3. **ATP and ADP** need special transporter to allow ADP in and ATP out of the mitochondria

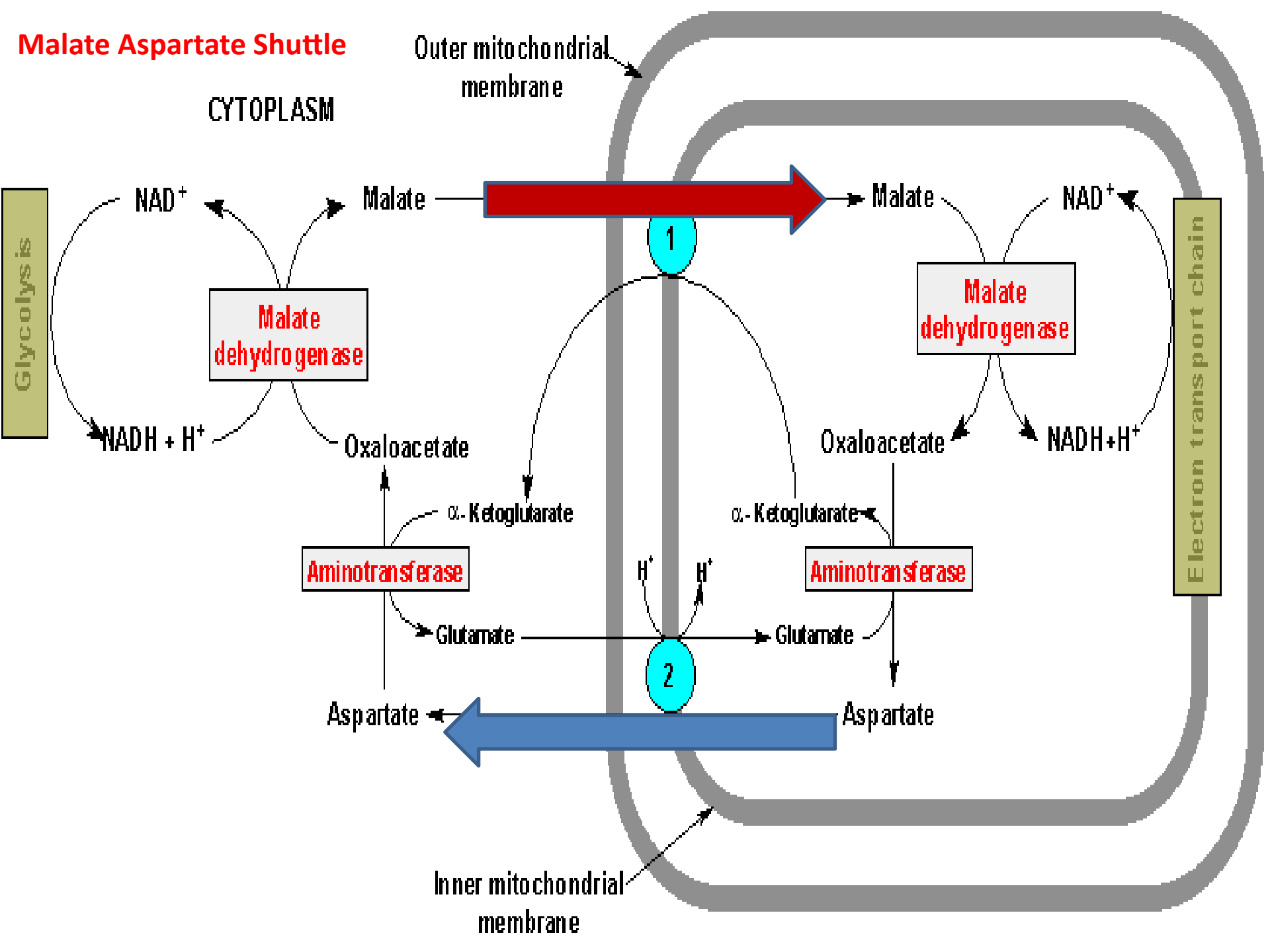


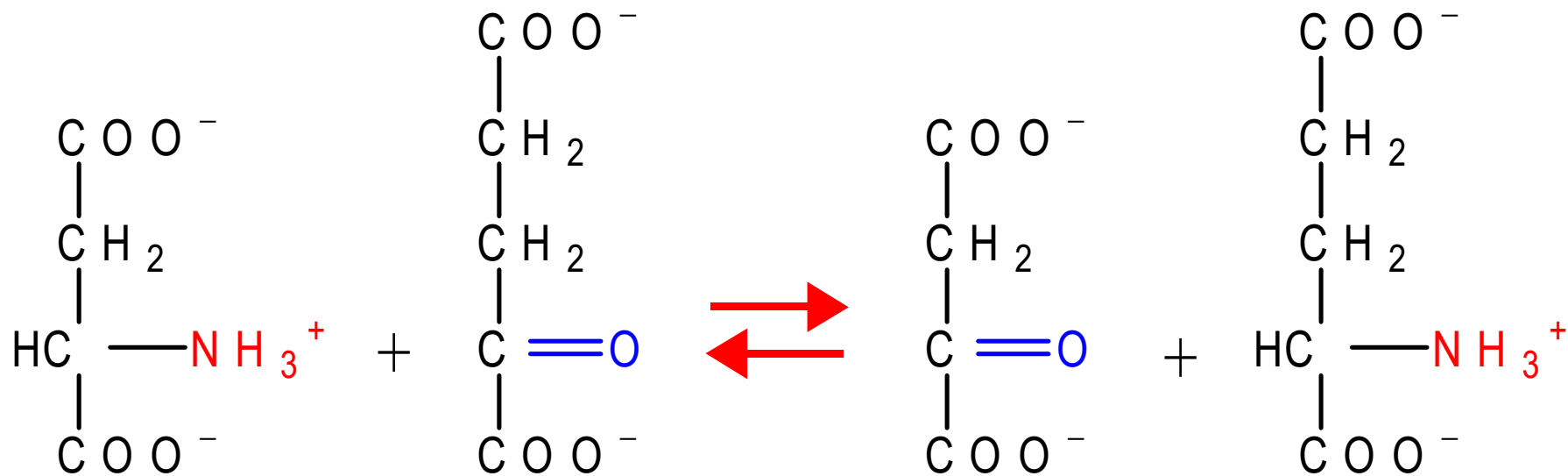
Glycerol 3 Phosphate Shuttle

CYTOPLASM



Malate Aspartate Shuttle





aspartate α-ketoglutarate oxaloacetate glutamate

Aminotransferase (Transaminase)

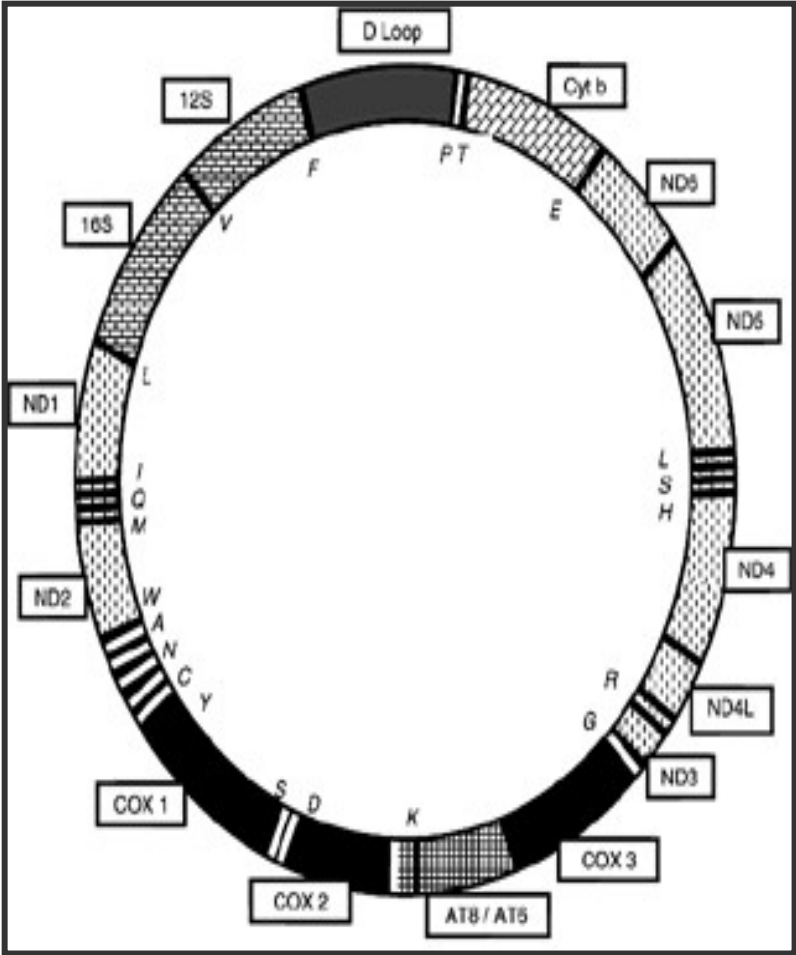
Mitochondrial DNA

Circular DNA

Derived from mother's ovum

It contains 37 genes

13 proteins of ETC
22 tRNA
2rRNA



Mitochondrial abnormalities

1. Mitochondrial Myopathies; it is due to defective production of energy leading to muscle cramping, weakness and severe fatigue.
2. Leber's hereditary optic neuropathy:
bilateral loss of vision and damage of optic nerve .

- How does **atractyloside** affect mitochondrial respiration? ...
- If you isolate mitochondria and place them in buffer with **a low pH**, they begin to manufacture ATP. Why?